

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

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Abstract

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account in an emotional Context Maintenance and Retrieval (eCMR) model, in which context reinstatement, competitor encoding, and retrieval control produce the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state binds task representations to overlapping context. When later retrieval contexts cue both film representations and task competitors, uncontrolled retrieval from context cues is less likely to sample trauma-film items in intrusion-like retrieval, even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals and control processes reinstate film context and prioritize film items over competitors for retrieval; film-specific cues can also steer retrieval toward film context across test formats. Together, the simulations reproduce selective impairment of intrusion-like retrieval and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both uncontrolled and controlled access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding or after a later film reminder, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Asselbergs et al., 2023](#); [Deepprose et al., 2012](#); [Holmes et al., 2009, 2010](#); [James et al., 2015](#); [Kessler et al., 2020](#); [Lau-Zhu et al., 2019](#)). Even for memories of real-life traumatic

events, including motor vehicle accidents, emergency caesarean section, and war-related trauma, the same post-reminder intervention can reduce intrusions while leaving deliberate recall relatively intact ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent interpretation treats the selective interference effect as evidence for separable memory representations ([Brewin et al., 1996](#); [Brewin, 2014](#)). Under this dual-representation account, traumatic experience produces distinct sensory and contextual memory traces — the first supporting involuntary re-experiencing, the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). A visuospatial interference task competes for the perceptual processing resources needed to consolidate the sensory trace, reducing later intrusions while leaving the contextual trace and the voluntary memory it supports relatively intact ([Holmes et al., 2009](#); [James et al., 2015](#)). When the intervention follows a reminder at longer delays, the account invokes reconsolidation: the reminder reactivates the sensory trace, and the interfering task disrupts its restabilization during a time-limited window ([James et al., 2015](#); [Kindt et al., 2009](#)). The account thus ties selective interference to separate memory representations, modality-specific disruption of one trace, and a discrete reconsolidation window.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation, the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal within a retrieved-context framework drawing on the emotional Context Maintenance and Retrieval model family, including eCMR and CMR3 (Cohen & Kahana, 2022; Talmi et al., 2019). Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009). Emotional extensions add a source-context stream for emotional and neutral states and an emotion-dependent binding strength, allowing the same architecture to capture how emotional content changes the organization and success of recall (Cohen & Kahana, 2022; Talmi et al., 2019). Variants also model strategic control over memory search, allowing the model to capture how retrieval goals and task demands shape recall (Cohen & Kahana, 2022; Lohnas et al., 2015; Sederberg et al., 2008). Together, these commitments make the framework a principled setting for testing whether selective interference can arise within a single emotional episodic-memory system.

Consolidation and reconsolidation accounts locate selective interference in a change to the availability of the film representation after encoding or reminder reactivation. The retrieved-context account developed here instead treats interference as a change in the competitive retrieval environment around the film. Post-film or post-reminder activity can add competitors near the film in context, while deliberate recall can partially protect access by guiding retrieval toward the film. On this view, the intrusion-voluntary-memory dissociation reflects different retrieval states applied to shared memory associations.

Our simulations evaluate the account at the level where a retrieved-context model makes its distinctive predictions: the distribution and organization of recall across the film, interference, and delay phases of a trauma-film paradigm. They first demonstrate the core selective-interference dissociation without separately vulnerable representations: an interference task can reduce intrusion-like retrieval more than deliberate recall, even though the same stored film episode supports both. They then distinguish the control mechanisms that protect deliberate recall, showing how a more film-directed retrieval state and a sharper decision rule produce different patterns of impairment in both recall rates and recall organization. The next simulations separate

the contributions of competitor strength and contextual overlap, clarifying when an interference task becomes an effective competitor for film retrieval and how reminder-driven reinstatement can support delayed interference. Follow-up simulations use the same framework to interpret heterogeneity across empirical paradigms. They show that film-related cues during intrusion monitoring can narrow the intrusion-voluntary-memory dissociation by making intrusion-like retrieval less purely unguided. They also show that item-specific or film-specific probes can make voluntary access less vulnerable to interference than tests that rely on broad context-to-item search, because broad search leaves film items competing more directly with task competitors. The result is a mechanistic alternative to separately vulnerable memory representations and a set of specific hypotheses about when selective interference should appear and how it should vary across paradigms.

A Retrieved-Context Account of Selective Interference

The account developed here builds on a contextual binding theory of episodic memory: experienced events are encoded by binding item representations to the current state of a continuously evolving context ([Howard & Kahana, 2002](#); [Yonelinas et al., 2019](#)). Context encompasses temporal, source, and other features that can change at different rates across experience ([Yonelinas et al., 2019](#)). Events encoded in similar context states compete during retrieval, so selective interference can arise when task events are encoded near film events in context ([Mensink & Raaijmakers, 1989](#); [Yonelinas et al., 2019](#)).

Retrieved-context theory formalizes this binding framework with two associative pathways ([Howard & Kahana, 2002](#); [Polyn et al., 2009](#)). Context-to-item associations allow the current context state to cue all events encoded in similar states, producing competitive retrieval in which events vie for recall based on the strength of their contextual match. Item-to-context associations allow a recalled or re-encountered event to reinstate the context in which it was originally experienced, a mechanism absent from earlier context theories. This reinstatement mechanism creates a bidirectional loop: retrieving an event drives context back toward the state present at encoding, which in turn cues neighboring events, producing the temporal contiguity effect

([Kahana, 1996](#); [Sederberg et al., 2008](#)).

The critical property for the present account is that context-to-item retrieval is competitive: any event encoded in the current contextual neighborhood is a candidate, and adding new events to that neighborhood increases retrieval competition. In the trauma-film paradigm, intrusive memories arise through this pathway as ongoing context drifts near states associated with the film, cueing film events without intention. When a participant performs a visuospatial task such as Tetris immediately after viewing a distressing film, the events the task produces are encoded in temporal context that overlaps substantially with the film's contextual neighborhood. When context later drifts near the film region, interference events compete alongside film events, reducing the probability that any single film event is retrieved as an intrusion. The strength of this interference depends on how many competitors are encoded, how strongly they are bound, and how much their associated temporal or source/emotional context overlaps with the film's. Perceptual modality can matter in this framework, but only insofar as it changes competitor strength or overlap with the film-related context.

The item-to-context pathway extends this account to post-reminder interference. In the trauma-film paradigm, presenting a film clip as a reminder retrieves its associated film context via item-to-context retrieval, pulling the current context back toward the state present during the original viewing ([James et al., 2015](#); [Kessler et al., 2020](#)). A visuospatial task performed in this reinstated state encodes competitors that overlap with the film context, enabling interference through the same contextual competition as the immediate case. Interference effectiveness depends on two continuous variables: how strongly the reminder reinstates film context, and how much subsequent experience has caused the reinstated context to drift before the interference task begins. In the model, delayed interference is therefore governed by graded reinstatement and overlap before competitor encoding.

Deliberate recall of the film presents a subtler case. Like intrusions, it uses context-to-item retrieval and is therefore exposed to competitor interference. Two control mechanisms can partially offset this vulnerability. Reinstating the film's starting context as the initial retrieval cue

biases recall toward the film region and away from the interference neighborhood; sharpening competition among candidates amplifies strongly associated film events relative to weakly matched competitors. Together, these mechanisms produce graded protection, with the degree of sparing depending on how much control the participant exerts.

Item-to-context retrieval is less exposed to contextual competition than context-to-item retrieval. For example, in recognition, the test supplies a film item as the retrieval probe rather than asking the current context to sample among film and task competitors. The probe retrieves its own associated context through item-to-context associations formed during film encoding. Interference items encoded later can add competitors to context-to-item retrieval, but they do not directly overwrite the film item's own item-to-context association. Recognition-like access therefore leaves little room for the contextual-competition mechanism that reduces intrusion-like recall. This protection is different from the graded protection available to deliberate free recall, which still depends on context-to-item retrieval and must use control to favor film items over competitors.

These mechanisms reproduce several core findings from the episodic memory literature. In free recall, the context-reinstatement loop accounts for temporal contiguity (the tendency to recall temporally neighboring events in sequence) and for the persistence of this pattern across long filled delays ([Howard & Kahana, 2002](#); [Kahana, 1996](#); [Sederberg et al., 2008](#)). Memories from recent but irrelevant experiences also intrude during recall of a target episode, with a frequency that decays gradually over intervening time ([Howard et al., 2008](#); [Lohnas et al., 2015](#); [Zaromb et al., 2006](#)). Emotional extensions of CMR add a source-context stream that can distinguish items encoded in emotional versus neutral states ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)). In the present source-only variant, this stream is used to represent whether film and task items share or differ in source/emotional context.

The resulting account has three operative ingredients. Intrusion-like and deliberate recall operate over the same item-context associations, but they begin from different retrieval states and differ in how strongly retrieval is constrained toward the film. Interference strength is determined

by the number and strength of task competitors and by their overlap with film-related temporal and source/emotional context. Post-reminder effects arise when a reminder reinstates film context before competitor encoding. These commitments define the model components specified next: phase-specific encoding, reminder-driven reinstatement, and retrieval-mode settings.

Model and Paradigm

Representations and Memory Architecture

The model represents each study event as an orthonormal item vector f_i . The selective-interference paradigm presents items across a fixed phase sequence: film, break, reminder, interference, filler, and retrieval. Film items are the target items whose later access indexes intrusion-like or voluntary memory. Interference items are later task events that can become competitors for film retrieval. Each item also carries a source flag $e_i \in \{0, 1\}$ indicating whether it belongs to the source/emotional class manipulated in a given simulation, and a target flag $t_i \in \{0, 1\}$ indicating whether it is a film item.

Two context streams evolve over the experiment. Temporal context c^T is a unit-length vector with a start-of-list unit and one unit for each item. Source context c^S is a unit-length vector with start, emotional/source, and neutral/source dimensions. Both context streams begin in their start states. As items are encoded or retrieved, context drifts toward the context associated with the current item.

Four associative matrices link items to contexts. $M^{FC,T}$ maps item features to temporal context and supports item-to-context reinstatement. $M^{CF,T}$ maps temporal context to item features and supports context-to-item competition. $M^{FC,S}$ and $M^{CF,S}$ provide the corresponding item-to-source-context and source-context-to-item mappings. The source pathway is the source-only eCMR extension used here: it can add source-based support for retrieval without making emotional/source status a hidden modulation of temporal learning.

Table 1 lists the model structures and parameters used throughout the simulations.

Pre-Experimental Associations

Pre-experimental initialization establishes baseline associations before the experiment.

For the temporal pathway:

$$M_{pre}^{FC,T}(i, j) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$M_{pre}^{CF,T}(j, i) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases}$$

The start context has no pre-experimental context-to-item support. For item-specific context units, δ gives each item self-support and α gives all other items shared support. This support is explicit so that simulations can test whether selective-interference effects depend on learned contextual competition rather than hidden background support.

For the source pathway, define each item's source vector as:

$$u_i = \begin{bmatrix} 0 \\ e_i \\ 1 - e_i \end{bmatrix}$$

$$M_{pre}^{FC,S}(i, k) = (1 - \gamma)u_{ik}$$

$$M_{pre}^{CF,S}(k, i) = 0$$

The source pathway therefore begins with item-to-source associations but no source-to-item competition. Source-based competition is learned during encoding and is controlled by the source learning scale.

Encoding Across Phases

The film, break, interference, and filler phases use the same encoding operation with phase-specific rates. Let p index the current phase and let β_p and λ_p denote that phase's temporal drift rate and context-to-item learning scale. The film phase encodes target items with baseline rates. The break and filler phases add intervening experience. The interference phase encodes task items whose effect depends on their strength and their overlap with film context.

For item i , temporal context input is retrieved by probing item-to-context memory:

$$c_i^{IN,T} = M^{FC,T} f_i$$

The model normalizes the input to unit length. Temporal context then drifts toward the normalized input:

$$c_i^T = \rho_i^T c_{i-1}^T + \beta_p c_i^{IN,T}$$

$$\rho_i^T = \sqrt{1 + \beta_p^2 \left[\left(c_{i-1}^T \cdot c_i^{IN,T} \right)^2 - 1 \right]} - \beta_p \left(c_{i-1}^T \cdot c_i^{IN,T} \right)$$

The ρ_i^T term preserves unit length. The drift rate β_p controls how far the current phase moves context away from the prior phase. In selective-interference simulations, this rate helps determine whether interference items are encoded near or far from the film context.

Feature-to-context associations update by an outer product after context updating:

$$\Delta M^{FC,T} = \gamma f_i (c_i^T)^\top$$

A primacy function weights temporal context-to-item learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1$$

$$\Delta M^{CF,T} = \lambda_p \phi_i c_i^T f_i^\top$$

Here, λ_p is the phase-specific MCF learning scale. Increasing λ_p during interference makes task items stronger competitors for later context-to-item retrieval. The primary source-only eCMR variant does not allow source/emotional status to modulate temporal context-to-item learning; temporal learning remains primacy-driven unless a later sensitivity analysis explicitly turns on broad temporal modulation.

Source/Emotional Context

Source context updates in parallel with temporal context. For item i , source-context input is retrieved by probing $M^{FC,S}$:

$$c_i^{IN,S} = M^{FC,S} f_i$$

Source context then integrates this input:

$$c_i^S = \rho_i^S c_{i-1}^S + \beta_{enc}^S c_i^{IN,S}$$

Source feature-to-context learning mirrors the temporal pathway:

$$\Delta M^{FC,S} = \gamma f_i (c_i^S)^\top$$

Source context-to-item learning is gated by the source flag:

$$\Delta M^{CF,S} = \lambda_p \phi_i \omega_S e_i c_i^S f_i^\top$$

The parameter ω_S controls source-context-to-item learning for source/emotional items. When $\omega_S = 0$ or when all items are neutral, the source pathway does not contribute source-to-item competition. When film and interference items share source/emotional status, the source pathway can increase target-competitor overlap beyond temporal proximity alone.

Each phase applies multiplicative scale factors to the base encoding rates. The film phase uses base rates directly. Subsequent phases modulate drift rate, which controls context proximity to preceding items, and M^{CF} learning scale, which controls competitor encoding strength.

The interference scales s_{int}^{β} and s_{int}^{ϕ} are central to the simulations. The drift scale controls how much temporal context changes during interference encoding. The MCF scale controls competitor encoding strength: stronger competitors capture more retrieval probability from film targets. Together, drift and learning scales define whether interference is merely added experience or a set of strong competitors encoded into film-related context.

Reminder Reinstatement

In the trauma-film paradigm, a film clip presented as a reminder between the break and interference phases serves to reinstate the film's encoding context. The model implements this in two steps. First, both context representations drift toward their respective start-of-list states:

$$c_{\text{rem}}^T = \rho c^T + \beta_{\text{rem,start}} c_0^T$$

$$c_{\text{rem}}^S = \rho_S c^S + \beta_{\text{rem,start}} c_0^S$$

Here, $\beta_{\text{rem,start}} = s_{\text{rem,start}}^{\beta} \cdot \beta_{\text{start}}$ sets how strongly context shifts toward the pre-experimental state before the reminder arrives. This partial reinstatement moves context back toward the general region of the film's temporal neighborhood.

Second, each reminder item reinstates its specific temporal and source context through the same item-to-context pathway used during recall:

$$c_{\text{cue}}^{IN,T} = M^{FC,T} f_{\text{cue}}$$

$$c^T \leftarrow \rho_T c^T + \beta_{\text{rem}} c_{\text{cue}}^{IN,T}$$

$$c_{\text{cue}}^{IN,S} = M^{FC,S} f_{\text{cue}}$$

$$c^S \leftarrow \rho_S c^S + \beta_{\text{rem}} c_{\text{cue}}^{IN,S}$$

Here, $\beta_{\text{rem}} = s_{\text{rem}}^{\beta} \cdot \beta_{\text{enc}}$ controls how strongly the reminder pulls context toward the film's encoding state. Critically, reminders reinstate context without forming new associations: $M^{FC,T}$, $M^{CF,T}$, $M^{FC,S}$, and $M^{CF,S}$ remain unchanged. This distinction mirrors the clinical logic of a reminder cue — it reactivates the memory without adding new content.

The reminder mechanism has a direct consequence for interference. By pulling context back toward the film neighborhood, the reminder makes interference items encoded immediately afterward share temporal and potentially source context with the film. Stronger reinstatement (s_{rem}^{β} closer to 1) produces more overlap, and hence more competition, between film targets and interference competitors. Delayed-interference simulations therefore manipulate reminder reinstatement as a graded overlap mechanism rather than as an all-or-none reactivation window.

Retrieval and Termination

Context-to-item retrieval underlies both intrusion-like retrieval and deliberate free recall. At the start of retrieval, both context streams drift toward their initial states:

$$c_{\text{start}}^T = \rho c_N^T + \beta_{\text{start}} c_0^T$$

$$c_{\text{start}}^S = \rho_S c_N^S + \beta_{\text{start}} c_0^S$$

Here, β_{start} controls how much start-of-list context is reinstated at retrieval onset. The current temporal and source contexts then cue all recallable items through $M^{CF,T}$ and $M^{CF,S}$:

$$a_i^T = [M^{CF,T} c^T]_i r_i$$

$$a_i^S = [M^{CF,S} c^S]_i r_i$$

$$a_i = a_i^T + a_i^S$$

Here, r_i is a binary mask indicating whether item i is currently recallable. Because all recallable items contribute to the denominator, encoding additional competitors in contextually similar states dilutes each film item's share of retrieval probability. The choice sensitivity τ power-scales the combined activation vector, with a small lower bound ϵ applied for numerical stability:

$$A_i = (a_i^\tau + \epsilon) r_i$$

The probability of recalling item i then follows a Luce choice rule:

$$P(i \mid \text{recall}) = \frac{A_i}{\sum_k A_k}$$

At each retrieval step, the model may terminate with probability p_{stop} or recall an item with probability $(1 - p_{\text{stop}}) \cdot P(i \mid \text{recall})$. Termination probability grows with recall position:

$$p_{\text{stop}}(j) = \theta_s e^{\theta_r \cdot j}$$

Here, j is the number of items already recalled, θ_s sets the baseline termination probability, and θ_r controls its growth. When an item is recalled, both context streams reinstate its associated context:

$$c^T \leftarrow \rho_T c^T + \kappa_i \beta_{rec} M^{FC,T} f_i$$

$$c^S \leftarrow \rho_S c^S + \kappa_i \beta_{rec}^S M^{FC,S} f_i$$

The target-monitoring scale κ_i equals 1 for film items and can be reduced for non-film competitors. This target monitoring setting represents a retrieval policy in which rejected competitors have less influence on the next context state. This reinstatement drives context toward the recalled item's encoding neighborhood, biasing subsequent retrievals toward temporally and

emotionally proximate items. This mechanism produces the temporal organization expected of retrieved-context models.

Retrieval Configurations

The simulations vary retrieval configuration without changing the stored film and task episodes. Intrusion-like retrieval is modeled as relatively broad context-to-item sampling from the current retrieval context. Deliberate recall uses the same context-to-item pathway, but with stronger film-directed start reinstatement, sharper selection among candidates, and optionally reduced context updating after rejected non-film competitors. These settings do not create a separate voluntary-memory store; they change the retrieval state and selection policy applied to the same item-context associations.

Probe-driven access is treated as a boundary case rather than as a categorical immunity claim. When a film item is supplied as a probe, it can retrieve its own associated context through item-to-context memory:

$$c_i^{IN,T} = M^{FC,T} f_i$$

This route supplies more item-specific information than broad context-to-item search. It should therefore be less exposed to competitors encoded near the film context, although the manuscript treats this as a boundary case to be interpreted alongside the task design rather than as an absolute recognition result.

Parameters and Simulation Factors

The base parameter vector is anchored by external eCMR fits rather than fit to the selective-interference simulations themselves. For these simulations, the anchor is a pooled source-only eCMR fit to the Ukraine war photographs free-recall data ([Dupertuys et al., 2026](#)). The selective-interference simulations then manipulate paradigm-level scale factors that determine phase drift, competitor learning, source/emotional overlap, reminder reinstatement, and retrieval control. This separation keeps fitted memory-search parameters distinct from the theoretical manipulations used to create selective interference.

Table 3 lists the main scale factors and their theoretical roles.

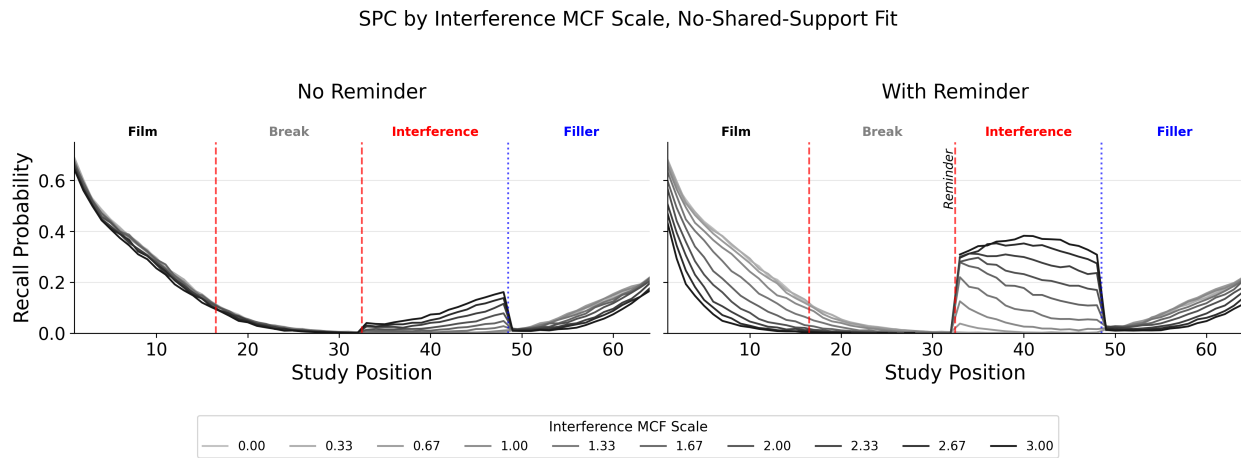
The paradigm geometry mirrors trauma-film selective-interference designs at an abstract level. Each simulated trial includes film, optional break, reminder, interference, and filler phases, with phase lengths and phase-specific scaling set by the simulation factors in Table 3. The main simulation outputs track which phase each recalled item came from, how often film items are recalled, and how those quantities change across interference and retrieval-control manipulations. Because retrieval proceeds as a sequence of competitive samples terminated by a growing stop probability, the expected number of film items recalled depends on how much retrieval probability competitors capture at each step. These dependent measures provide a bridge between the model's formal predictions and the empirical phenomena the simulations target.

Simulations

The first rendered sweep included here orients the reader to the simulated phase structure before the later sections develop the final manuscript contrasts. These plots are diagnostic outputs from the interference-strength sweep, not stand-alone evidence for the full intrusion-voluntary-memory dissociation.

Figure 1

Phase-resolved recall in the interference-strength sweep. Serial-position curves show recall probability across the film, break, interference, and filler phases of the simulated paradigm, separately for no-reminder and reminder conditions. Darker lines indicate stronger context-to-item learning for interference-task items. The figure shows how the interference manipulation changes the distribution of retrieval across encoded phases, rather than only changing total film recall.



Simulation 1: Core Selective-Interference Dissociation

Premise: The basic selective-interference result is a 2 x 2 contrast between interference condition and retrieval configuration. The section asks whether the same stored film episode and the same post-film interference manipulation can produce larger impairment in intrusion-like retrieval than in deliberate recall.

Intended text take-away: This section establishes the core sufficiency result. The model can reproduce the qualitative intrusion-voluntary-memory dissociation without separately weakening film items for one measure. Deliberate recall may still be impaired; the claim is relative protection, not immunity.

Simulation 2: Control Mechanisms Underlying Deliberate-Recall Protection

Premise: Deliberate recall is not categorically protected. Its relative sparing depends on control operations that make retrieval more film-directed.

Intended text take-away: This section earns the “relatively intact voluntary memory” claim. Sparing is a graded consequence of retrieval control rather than a property attached to voluntary tests as a category. Weakening these control operations should make deliberate recall resemble intrusion-like retrieval more closely.

Simulation 3: Competitor Strength, Contextual Overlap, and Emotional-Context Amplification

Premise: An interference task matters when it encodes competitors that are both strong enough and contextually close enough to film items. This section asks what makes a task Tetris-like in model terms.

Intended text take-away: This section explains what makes interference effective. The account should not reduce to generic distraction, list length, or task difficulty. Competitor strength matters because competitors are retrievable from film-related context, and shared source/emotional context is relevant only insofar as it increases that competition.

Simulation 4: Reminder Reinstatement and Delayed Interference

Premise: The delayed-reminder case applies the overlap mechanism after context has drifted away from the film. A reminder matters because it reinstates film context before competitor encoding.

Intended text take-away: This section connects the model to post-reminder interventions. The reminder is not treated as a special storage-disruption device. It restores the contextual condition required for later competitor encoding to affect film-item retrieval.

Simulation 5: Heterogeneity Across Recall-Like Test Formats

Premise: Intrusion diaries, vigilance-style monitoring, free recall, and cued recall differ in how much they rely on broad context-to-item search versus film-directed retrieval support. This section maps the model’s retrieval-control mechanisms onto variation across recall-like empirical measures.

Intended text take-away: This section interprets heterogeneity across empirical paradigms without treating intrusion and voluntary memory as fixed computational kinds. The model

predicts a continuum of vulnerability across recall-like tests depending on how much the task supports film-directed retrieval.

Simulation 6: Probe-Driven Access as a Boundary Case

Premise: Item-specific probes are a limiting case of voluntary access because they begin closer to the target item than broad recall does. This section preserves the useful part of the old recognition result without treating probe-driven access as categorically protected.

Intended text take-away: This section handles film-specific probes and recognition-like measures as a boundary case. The claim is that probe-driven access reduces exposure to competitor competition, not that recognition escapes the mechanism in every design or that all voluntary memory has the same protection.

General Discussion

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Table 1

Model structures and parameters. Base parameters are anchored by pooled source-only eCMR fits to the Ukraine war photographs free-recall data (Dupertuys et al., 2026).

Symbol	Name	Description
c^T	temporal context	Recency-weighted context stream with start and item units
c^S	source context	Source/emotional context stream with start, emotional/source, and neutral/source units
f_i	item features	Orthonormal representation of item i
e_i	source flag	1 for source/emotional items, 0 for neutral/source items
t_i	target flag	1 for film items, 0 for non-film items
$M^{FC,T}, M^{CF,T}$	temporal memories	Item $\leftrightarrow$ temporal-context associations
$M^{FC,S}, M^{CF,S}$	source memories	Item $\leftrightarrow$ source-context associations
β_{enc}	encoding drift rate	Temporal context drift during study
β_{start}	start drift rate	Start-of-list reinstatement at recall onset
β_{rec}	recall drift rate	Context drift during recall
β_{enc}^S	source encoding drift rate	Source-context drift during encoding
β_{rec}^S	source recall drift rate	Source-context drift during recall

Table 2

Phase-specific encoding rates. Drift scales multiply β_{enc} and are clipped to $[0, 1]$.

Phase	Drift rate	M^{CF} learning scale	Paradigm role
Film	β_{enc}	1	Target encoding
Break	$s_{break}^{\beta} \cdot \beta_{enc}$	s_{break}^{ϕ}	Retention interval
Interference	$s_{int}^{\beta} \cdot \beta_{enc}$	s_{int}^{ϕ}	Competitor encoding
Filler	$s_{fill}^{\beta} \cdot \beta_{enc}$	s_{fill}^{ϕ}	Pre-recall buffer

Table 3

Scale factors and their simulation connections. Drift scales are clipped to [0, 1] after multiplication.

Scale factor	Base parameter or target	Role	Simulation connection
s_{break}^{β}	β_{enc}	Break-phase drift	Delay and context separation
s_{break}^{ϕ}	ϕ_i	Break-phase competitor learning	Break-task control
$s_{\text{rem,start}}^{\beta}$	β_{start}	Start-state shift before reminder	Reminder reinstatement
s_{rem}^{β}	β_{enc}	Film-item reminder drift	Reminder reinstatement
s_{int}^{β}	β_{enc}	Interference-phase drift	Target-competitor contextual overlap
s_{int}^{ϕ}	ϕ_i	Interference-phase MCF learning	Competitor strength
s_{fill}^{β}	β_{enc}	Filler-phase drift	Post-interference buffer
s_{fill}^{ϕ}	ϕ_i	Filler-phase MCF learning	Filler-task control
s_{start}^{β}	β_{start}	Retrieval start reinstatement	Film-directed recall control
s^{τ}	τ	Retrieval selection sharpness	Recall control
ω_S	source MCF learning	Source/emotional learning strength	Source/emotional overlap
$s_{\text{enc}}^{\beta,S}$	β_{enc}^S	Source-context encoding drift	Source/emotional overlap
$s_{\text{rec}}^{\beta,S}$	β_{rec}^S	Source-context recall	Source-context